Branch: MCA (Data Science) Kargil	**Semester:** 2
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Subject Name: Technical Training - I Lab	**Subject Code**: 25CAP-652
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Experiment 10

Experiment Aim:

To implement Transaction Control Language (TCL) commands in PostgreSQL using COMMIT, ROLLBACK, and SAVEPOINT for maintaining data consistency and controlling transactions.

Tools used: PostgreSQL

Objectives:

- **Transaction Management:** To understand the use of TCL commands in PostgreSQL.
- **Rollback Operations:** To revert unwanted changes using ROLLBACK.
- **Savepoint Usage:** To create checkpoints within transactions using SAVEPOINT.
- **Data Integrity:** To maintain consistency during multiple database operations.

Theory:

Transaction Control Language (TCL) commands are used to manage transactions in a database. A transaction is a sequence of SQL operations performed as a single logical unit. PostgreSQL supports TCL commands such as COMMIT, ROLLBACK, and SAVEPOINT to ensure reliability, consistency, and recovery in case of errors.

Experiment Steps:

Step 1: Creating Table

Query:

```
CREATE TABLE Payroll (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2) CHECK(SALARY > 0)
);
```

Step 2: Inserting Sample Data

Query:

```
INSERT INTO Payroll VALUES
(1, 'Amit', 30000),
(2, 'Neha', 40000),
(3, 'Ravi', 50000);
```

SELECT * FROM Payroll;

Output:

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	50000.00

Step 3: Using ROLLBACK

Query:

```
-- using rollback
begin;

UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK;
```

UPDATE Payroll

SET salary = 1000

WHERE emp_id = 3;

SELECT * FROM Payroll

Output:

Data Output Messages Notifications

```
ERROR:  new row for relation "payroll" violates check constraint
"payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).

SQL state: 23514
Detail: Failing row contains (3, Ravi, -1000.00).
```

Data Output Messages Notifications

```
ROLLBACK

Query returned successfully in 131 msec.
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	1000.00

Step 4: Using SAVEPOINT

Query:

```
BEGIN;

UPDATE Payroll
SET salary = salary + 5000
WHERE emp_id = 1;

SAVEPOINT sp1;

UPDATE Payroll
SET salary = salary + 7000
```

```
WHERE emp_id = 2;

UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;

ROLLBACK TO sp1;

SELECT * FROM Payroll;

COMMIT;
```

Output:

```
Data Output   Messages   Notifications
UPDATE 1

Query returned successfully in 529 msec.
```

```
Data Output   Messages   Notifications
SAVEPOINT

Query returned successfully in 112 msec.
```

Data Output Messages Notifications

```
ERROR:  new row for relation "payroll" violates check constraint
"payroll_salary_check"
Failing row contains (3, Ravi, -1000.00).

SQL state: 23514
Detail: Failing row contains (3, Ravi, -1000.00).
```

Data Output Messages Notifications

```
ROLLBACK

Query returned successfully in 81 msec.
```

Data Output Messages Notifications

Showing rows: 1 to 3 Page No: 1 of 1

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	2	Neha	40000.00
2	3	Ravi	1000.00
3	1	Amit	35000.00

Outcomes:

- Students will be able to use TCL commands in PostgreSQL.
- Students will understand transaction rollback mechanisms.
- Students will learn to use SAVEPOINT for partial rollback.
- Students will maintain data consistency during transactions.